YEAR 10 SCIENCE (EXTENSION)
PHYSICAL SCIENCES
ASSIGNMENT - MOTION

NAME: SOLUTIONS **MARK:** — / 60

DUE DATE: ____________________

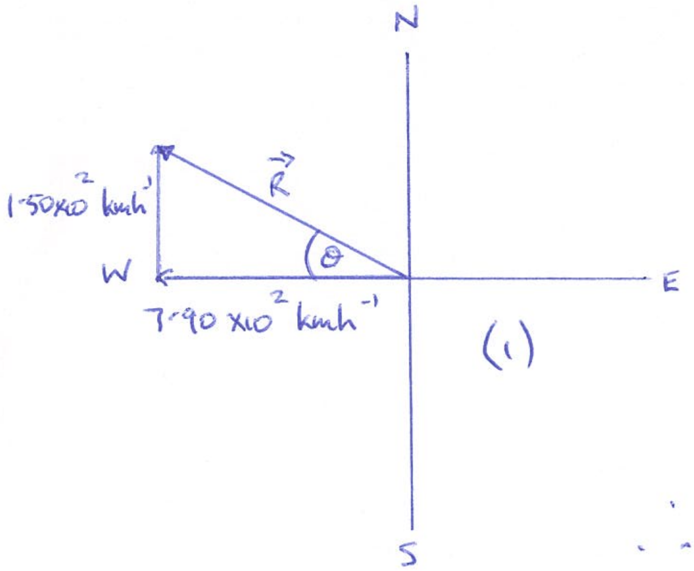
1. Perform the following conversions.

(a) 28.0 kmh^{-1} = 7·8 ms^{-1} (b) 3.7 tonnes = $3{\cdot}7 \times 10^{3}$ kg [1 mark each]

(c) 8.3 minutes = 498 s (d) 2.70 cm = 0·0270 m

(4)

2. A plane is flying at an altitude of 10,000 m at 7.90 x 10^2 kmh^{-1} W. There is a strong wind blowing from the south at 1.50 x 10^2 kmh^{-1}. Determine the plane's velocity over the ground. **(Include a vector diagram and give your answer in kmh^{-1}.)**



$$\vec{R} = \sqrt{(7{\cdot}90 \times 10^2)^2 + (1{\cdot}50 \times 10^2)^2}$$

$$= 8{\cdot}04 \times 10^2 \text{ kmh}^{-1} \quad (1)$$

$$\tan\theta = \frac{1{\cdot}50 \times 10^2}{7{\cdot}90 \times 10^2}$$

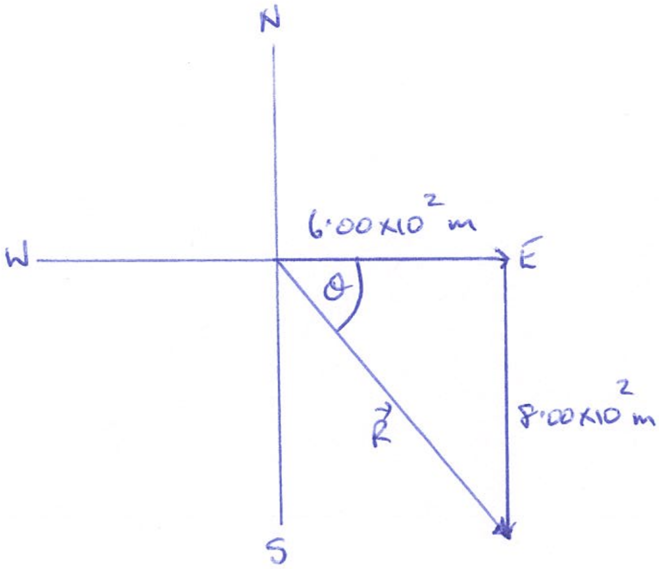
$$\Rightarrow \theta = 10{\cdot}8^\circ \quad (1)$$

$\therefore$ velocity = $8{\cdot}04 \times 10^2$ kmh^{-1} W10·8° N (1)

(4)

3. An athlete runs 6.00×10^2 m E in 95.0 s, then 8.00×10^2 m S in 2.00 minutes.

(a) Calculate her displacement.



$\vec{R} = \sqrt{(6{\cdot}00 \times 10^2)^2 + (8{\cdot}00 \times 10^2)^2}$

$= 1{\cdot}0 \times 10^3$ m (1)

$\tan\theta = \dfrac{8{\cdot}00 \times 10^2}{6{\cdot}00 \times 10^2}$

$\Rightarrow \theta = 53{\cdot}1°$ (1)

$\therefore$ $s = 1{\cdot}0 \times 10^3$ m E $53{\cdot}1°$ S

(1)

(3)

(b) Determine her average speed.

Speed = ?

$d = 1{\cdot}4 \times 10^3$ m (1)

$t = 215$ s

Speed $= \dfrac{d}{t}$

$= \dfrac{1{\cdot}4 \times 10^3}{215}$

$= \underline{6{\cdot}51 \text{ ms}^{-1}}$ (1)

(2)

(c) What is the athlete's average velocity?

$v = ?$

$s = 1{\cdot}0 \times 10^3$ m

$t = 215$ s

$v = \dfrac{s}{t}$

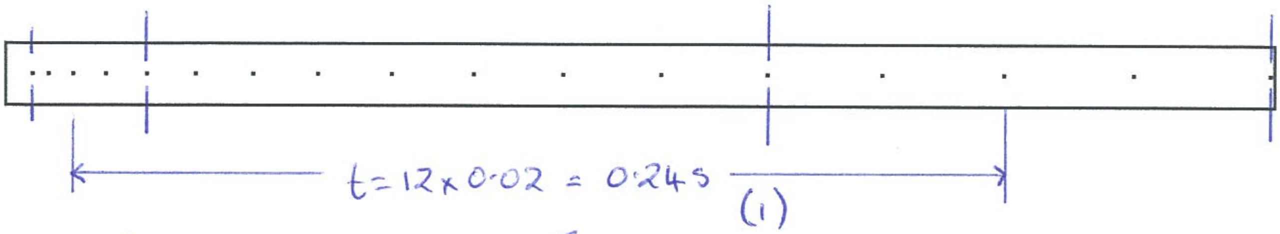
$= \dfrac{1{\cdot}0 \times 10^3}{215}$ (1)

$= 4{\cdot}65 \text{ ms}^{-1}$ E $53{\cdot}1°$ S

(1) (1)

(3)

4. In recording the motion of a toy car down an inclined track, a student produced the following tape using a ticker timer operating at 50 dots per second. Analyse the tape and determine the acceleration of the car. (Use the diagram below.)



$u = \frac{s}{t}$

$= \frac{1{\cdot}5 \times 10^{-2}}{0{\cdot}08}$

$= 0{\cdot}188\ ms^{-1}$ (1)

$v = \frac{s}{t}$

$= \frac{6{\cdot}7 \times 10^{-2}}{0{\cdot}08}$

$= 0{\cdot}838\ ms^{-1}$ (1)

$a = \frac{v-u}{t}$

$= \frac{0{\cdot}838 - 0{\cdot}188}{0{\cdot}24}$

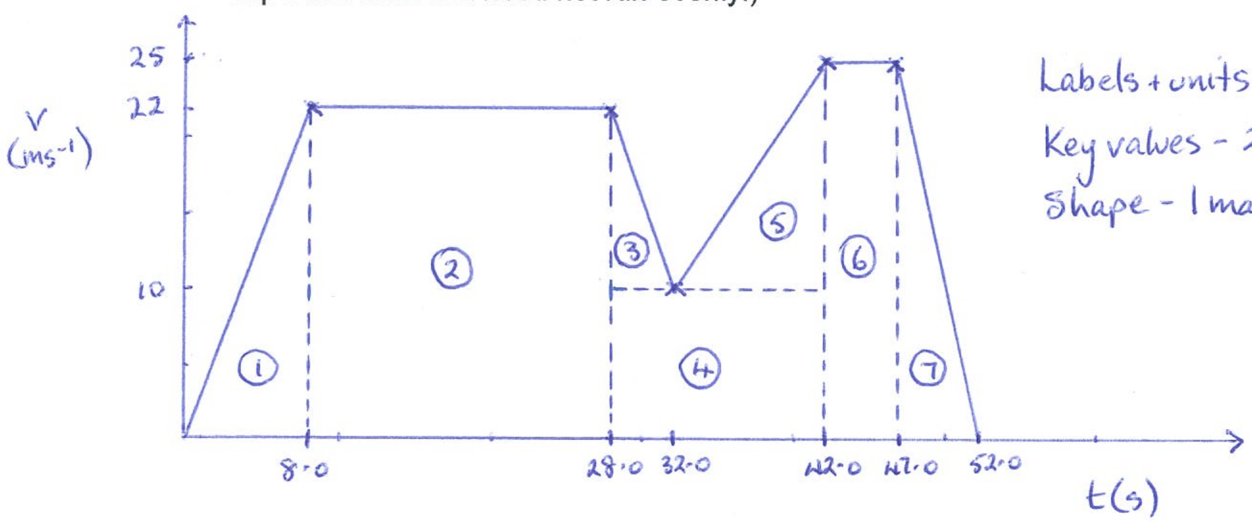
$= \underline{2{\cdot}71\ ms^{-2}}$ (1)

(4)

5. A car undergoes the following motion between three sets of traffic lights A, B and C.

The car accelerates uniformly from rest at light A and reaches 22.0 ms^{-1} after 8.00 s. It maintains this velocity for 20.0 s before slowing uniformly as it approaches lights B. The lights turn green so the car, having reduced its velocity to 10.0 ms^{-1} in 4.00 s, accelerates uniformly for 10.0 s to reach 25.0 ms^{-1}. It maintains this velocity for 5.00 s before uniformly decelerating to a stop in 5.00 s at lights C.

(a) Draw a velocity-time graph for the motion. (The scales only need to show the important data and need not run evenly.)



Labels + units - 1 mark

Key values - 2 marks

Shape - 1 mark

(4)

(b) Using the graph (or otherwise), calculate the size of the decelerations that occurred during the motion.

$$a_3 = \frac{(10.0 - 22.0)}{(32.0 - 28.0)} \quad (1)$$

$$= -3.00\,ms^{-2} \text{ backwards} \quad (1)$$

$$a_7 = \frac{(0 - 25.0)}{(52.0 - 47.0)} \quad (1)$$

$$= \underline{-5.00\,ms^{-2} \text{ backwards}} \quad (1)$$

(4)

(c) How far apart are lights A and C?

s = area under graph

$$= \tfrac{1}{2}(8.0)(22.0) + (20.0)(22.0) + \tfrac{1}{2}(4.0)(12.0) + (14.0)(10.0) + \tfrac{1}{2}(10.0)(15.0)$$

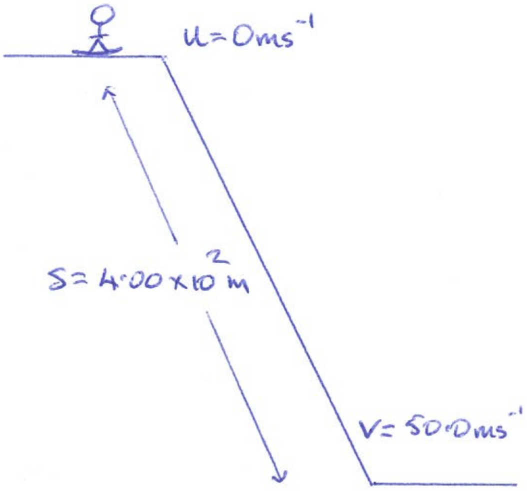
$$+ (5.0)(25.0) + \tfrac{1}{2}(5.0)(25.0) \quad (3)$$

$$= \underline{9.54 \times 10^{2}\,m} \quad (1)$$

(4)

6. A downhill skier of mass 80.0 kg started from rest on top of a mountain and accelerated smoothly down a uniform slope. After moving 4.00×10^2 m his velocity was 1.80×10^2 kmh^{-1}.

(a) Calculate the acceleration experienced by the skier down the slope.



Down the slope is +ve.

$v = 50.0ms^{-1}$

$u = 0.0ms^{-1}$

$a = ?$

$t = ?$

$s = 4.00 \times 10^2$ m

$v^2 = u^2 + 2as$ (1)

$\Rightarrow (50.0)^2 = 0 + 2a(4.00 \times 10^2)$ (1)

$\Rightarrow a = \dfrac{2.50 \times 10^3}{8.00 \times 10^2}$

$= \underline{3.12 ms^{-2} \text{ down the slope.}}$ (1)

(3)

(b) How long did it take for him to travel this far?

$v = u + at$ (1)

$\Rightarrow 50.0 = 0 + (3.12)t$ (1)

$\Rightarrow \underline{t = 16.0 s}$ (1)

(3)

7. A car engine supplies 4.50×10^3 N of force through the front wheels of a car and accelerates it uniformly from rest. The mass of the car is 1.40×10^3 kg.

(a) Calculate the acceleration of the car.

Forwards is +ve.

$\Sigma F = 4.50 \times 10^3$ N

$m = 1.40 \times 10^3$ kg

$a = ?$

$\Sigma F = ma$

$\Rightarrow a = \dfrac{\Sigma F}{m}$ (1)

$= \dfrac{4.50 \times 10^3}{1.40 \times 10^3}$ (1)

$= \underline{3.21\ ms^{-2}\ \text{forwards}}$ (1)

(3)

(b) Using your answer from part (a), calculate the velocity of the car after 5.0 s.

Forwards is +ve.

$v = ?$

$u = 0.0\ ms^{-1}$

$a = 3.21\ ms^{-2}$

$t = 5.0\ s$

$s = ?$

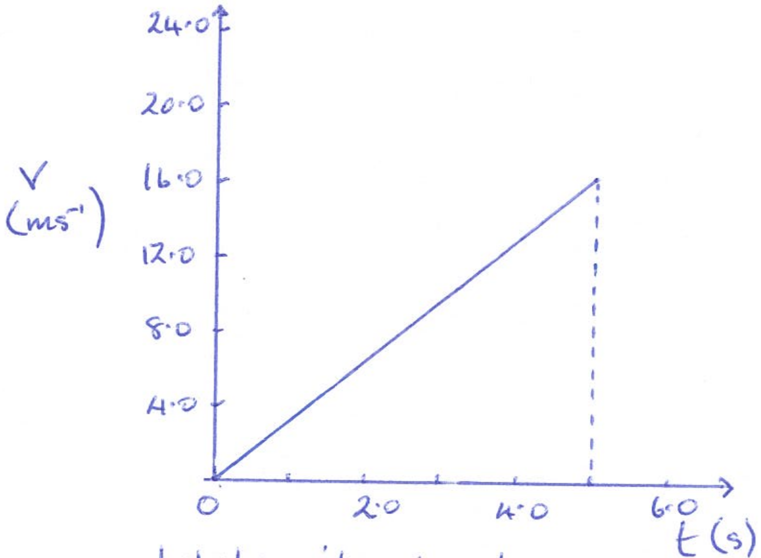
$v = u + at$ (1)

$= 0 + (3.21)(5.0)$ (1)

$= \underline{16.0\ ms^{-1}\ \text{forwards}}$ (1)

(3)

(c) Draw a velocity - time graph of the motion for 5.0 s and determine the displacement of the car.



s = area under graph

$= \frac{1}{2}(5.0)(16.0)$ (1)

$= \underline{40.0\ m\ \text{forwards}}$ (1)

Label + units - 1 mark

Plotting - 1 mark

(4)

8. A truck of mass 24.0 tonnes is moving at 80.0 kmh^{-1} along Roe Highway when the driver notices a small car that pulled out from a petrol station 60 m ahead. He hit the brakes sharply and just manages to avoid hitting the back of the car. The truck ended up moving at 30.0 kmh^{-1} and took 3.92 s to achieve this.

Calculate the average force exerted by the brakes of the truck.
(HINT: Calculate the acceleration first.)

Take forwards as +ve.

$v = 8.33\ ms^{-1}$

$u = 22.2\ ms^{-1}$

$a = ?$

$t = 3.92 s$

$s = ?$

$$a = \frac{v-u}{t} \quad (\tfrac{1}{2})$$

$$= \frac{8.33-22.2}{3.92} \quad (\tfrac{1}{2})$$

$$= -3.54\ ms^{-2}$$

$\therefore$ Deceleration $= 3.54\ ms^{-2}$ backwards (1)

$\Sigma F = ?$

$m = 24.0 \times 10^3\ kg$

$a = ?$

$$\Sigma F = ma \quad (\tfrac{1}{2})$$

$$= (24.0 \times 10^3)(-3.54) \quad (\tfrac{1}{2})$$

$$= -7.44 \times 10^4\ N$$

$\therefore$ $F = 8.50 \times 10^4$ N backwards. (1)

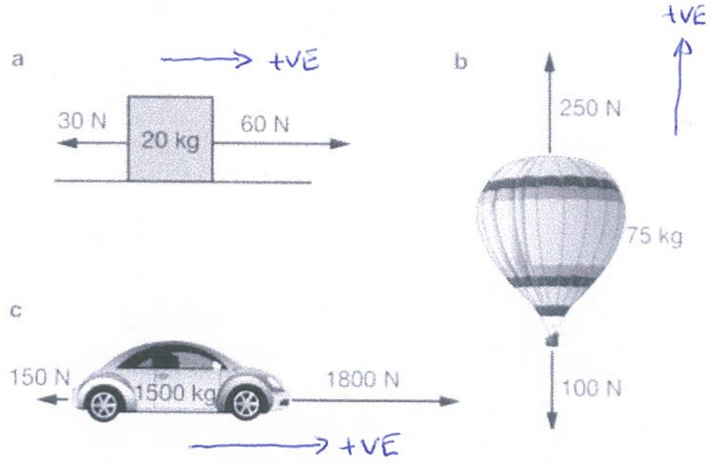
(4)

9. Explain why you "lurch forward" when standing on a bus or train and the vehicle brakes suddenly.

- You have inertia since you are moving. (1)
- As the vehicle slows, inertia carries you forward. (1)

(2)

10. Calculate the nett force and the acceleration for the following objects. Show your calculations.



(a) $\Sigma F = 60 - 30$

$= \underline{30\text{N to the right.}}$ (1)

$a = \frac{\Sigma F}{m}$

$= \frac{30}{20}$

$= \underline{1{\cdot}5\ \text{ms}^{-2}\ \text{to the right.}}$ (1)

(b) $\Sigma F = 250 - 100$

$= \underline{150\ \text{N up}}$ (1)

$a = \frac{\Sigma F}{m}$

$= \frac{150}{75}$

$= \underline{2{\cdot}0\ \text{ms}^{-2}\ \text{up}}$ (1)

(c) $\Sigma F = 1800 - 150$

$= \underline{1650\ \text{N forwards}}$ (1)

$a = \frac{\Sigma F}{m}$

$= \frac{1650}{1500}$

$= \underline{1{\cdot}1\ \text{ms}^{-2}\ \text{forwards.}}$ (1)

(6)